Thesis for Master's Degree

State-wise Safety in Autonomous Driving via Lagrangian-based Constrained Reinforcement Learning

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AI Graduate School

Gwangju Institute of Science and Technology

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석 사 학 위 논 문

라그랑지안 기반 제약 강화학습을 이용한 자율주행에서의 상태별 안전성 고려

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State-wise Safety in Autonomous Driving via Lagrangian-based Constrained Reinforcement Learning

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State-wise Safety in Autonomous Driving via Lagrangian-based Constrained Reinforcement Learning

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Dedicated to my family.

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Abstract

For the practical deployment of autonomous driving systems, high levels of safety and adaptability are essential. Deep Reinforcement Learning (DRL), which allows an agent to learn driving strategies through interaction with the environment, has attracted significant attention. However, DRL can raise safety concerns when applied in real-world scenarios due to challenges such as reward design and poor generalization to out-of-distribution situations. To address these issues, Constrained Reinforcement Learning (CRL) has been proposed, which seeks to balance safety and performance during policy learning. CRL is typically formulated using Constrained Markov Decision Processes (CMDPs), where constraints are defined as expected cumulative costs. Nevertheless, in autonomous driving, some constraints—such as collision avoidance—require immediate reactions. Thus, even when CMDP-based policies satisfy cumulative cost-based constraints, violations of instant constraints may still occur. In this paper, we propose a Lagrangian-based policy optimization method that estimates state-wise Lagrangian multipliers to precisely reflect constraints and consider state-level safety. The

proposed method is validated on a navigation task in the Safety Gym environment developed by OpenAI, where a vehicle must reach a designated goal location.

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국 문 요 약

자율주행 시스템의 실제 적용을 위해서는 높은 안정성과 적응성이 요구된다. 이를 위해 환경과의 상호작용을 통해 스스로 주행 전략을 학습하는 심층 강화 학습(Deep Reinforcement Learning, DRL)이 주목받고 있다. 그러나 심층 강화 학습은 보상 설계, 학습 분포 외의 상황에 대한 일반화 실패 등으로 인해 실제 적용 시 안전 문제가 발생할 수 있다. 이를 해결하고자 정책 학습 시 안정성과 성능 간의 균형을 도모하는 제약 강화 학습(Constrained Reinforcement Learning, CRL)이 제안되었다. 제약 강화 학습은 일반적으로 기댓값 기반 누적 비용 형태로 제약 조건을 정의하는 제약 마르코프 결정(Constrained Markov Decision Process, CMDP)에 기반하는데 자율주행에서는 충돌 회피와 같이 즉각적인 반응이 요구되는 제약 조건들이 존재하기 때문에, CMDP에서 정의한 누적 제약을 만족하도록 학습하더라도 이러한 즉각적인 제약을 위반하는 상황이 발생할 수 있다. 따라서 본 논문에서는 라그랑지안 기반 정책 최적화 방법을 통해, 상태별로 라그랑주 승수(Lagrange multiplier)를 추정하여 제약 조건을 정밀하게 반영함으로써 상태별 안정성을 고려하고자 한다. 제안한 방법은 OpenAI의 시뮬레이션 환경인 Safety Gym에서 차량이 지정된 목표 지점까지 이동하는 태스크를 통해 검증하였다.

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Chapter 1

Introduction

1.1 Introduction

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1.2 Motivation

insert your own materials [1].

	Overall classification results (%)		
Descriptors	KTH	UCF sports	TV human interaction
Trajectory	91.67	74.64	46.43
HOG	86.11	69.88	44.52
MBH	96.3	72.74	45.88
Combined	96.3	81.07	55.4

Table 1.1: An example table

Figure 1.1: The symbol of GIST.

Chapter 2

Preliminaries

2.1 Reinforcement Learning

Reinforcement Learning (RL) is a framework in which an agent interacts with an environment and learns a policy to maximize cumulative rewards. The agent observes the state of the environment and takes actions based on its policy, receiving rewards as feedback. This process is formalized as a Markov Decision Process (MDP), which provides a formal structure for decision making in RL. An MDP is defined by a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$, where $\mathcal{S}$ is the set of states, $\mathcal{A}$ is the set of actions, $\mathcal{P}$ is the state transition probability function, $\mathcal{R}$ is the reward function, and γ is the discount factor. The objective of RL is to find an optimal policy π^* that maximizes the expected cumulative reward, defined as:

$$\begin{aligned} \theta^* &= \arg\max J(\theta) \\ J(\theta) &= \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{t=0}^{T} r_t \right] \end{aligned} \tag{2.1}$$

The policy is assumed to be parameterized by θ and differentiable.

2.1.1 Policy Gradient Methods

Since the policy is differentiable, its gradient can be expressed using the likelihood ratio trick:

$$\begin{aligned}\nabla_\theta \pi_\theta(s,a) &= \pi_\theta(s,a)\frac{\nabla_\theta \pi_\theta(s,a)}{\pi_\theta(s,a)} \\ &= \pi_\theta(s,a)\nabla_\theta \log \pi_\theta(s,a)\end{aligned} \tag{2.2}$$

This term $\nabla_\theta \log \pi_\theta(s,a)$ is referred to as the score function. Although we previously defined the objective function $J(\theta)$ as the expected cumulative reward, we now consider a simplified case to facilitate explanation: a one-step MDP, in which the agent takes an action from the initial state, receives a reward, and the episode terminates immediately. Then, the objective function can be written as (d is the initial state distribution):

$$\begin{aligned}J(\theta) &= \mathbb{E}_{\pi_\theta}[r] \\ &= \sum_{s\in\mathcal{S}} d(s) \sum_{a\in\mathcal{A}} \pi_\theta(s,a) R_{s,a}\end{aligned} \tag{2.3}$$

The gradient of the objective function can be computed as:

$$\begin{aligned}\nabla_\theta J(\theta) &= \sum_{s\in\mathcal{S}} d(s) \sum_{a\in\mathcal{A}} \nabla_\theta \pi_\theta(s,a) R_{s,a} \\ &= \sum_{s\in\mathcal{S}} d(s) \sum_{a\in\mathcal{A}} \pi_\theta(s,a) \nabla_\theta \log \pi_\theta(s,a) R_{s,a} \\ &= \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a) r]\end{aligned} \tag{2.4}$$

The Policy Gradient Theorem generalizes this result to multi-step MDPs, where the objective function is defined as the expected cumulative reward over multiple time steps.

Theorem 2.1.1 (Policy Gradient Theorem)

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a) Q^{\pi_\theta}(s,a)] \tag{2.5}$$

In practice, the exact action-value function $Q^{\pi_\theta}(s,a)$ is typically unknown. Thus we use the empirical return(cumulative reward) as an unbiased estimate of the action-value function. This leads to the Monte Carlo Policy Gradient method, commonly known as REINFORCE algorithm. However, REINFORCE suffers from high variance in the gradient estimates due to its reliance on returns. A common approach to reducing variance is to use a critic that approximates the action-value function, i.e., $Q_W(s,a) \approx Q^{\pi_\theta}(s,a)$

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a) Q_W(s,a)] \tag{2.6}$$

This is the basic idea of Actor-Critic methods. Actor updates the policy parameters θ and a critic updates the value function parameters W. To further reduce variance, we can introduce a baseline function. Importantly, subtracting a baseline from the action-value function does not change the gradient estimate. Because the gradient of

the baseline function vanishes in expectation:

$$
\begin{aligned}
\mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a)B(s)] &= \sum_{s\in\mathcal{S}} d(s) \sum_{a\in\mathcal{A}} \pi_\theta(s,a)\nabla_\theta \log \pi_\theta(s,a)B(s) \\
&= \sum_{s\in\mathcal{S}} d(s) \sum_{a\in\mathcal{A}} \nabla_\theta \pi_\theta(s,a)B(s) \\
&= \sum_{s\in\mathcal{S}} d(s)B(s)\nabla_\theta \sum_{a\in\mathcal{A}} \pi_\theta(s,a) \\
&= 0
\end{aligned}
\tag{2.7}
$$

Therefore, subtracting a baseline function from the action-value function not only leaves the gradient estimate unchanged, but also significantly reduces its variance.

$$
\begin{aligned}
A^{\pi_\theta}(s,a) &= Q^{\pi_\theta}(s,a) - V^{\pi_\theta}((s) \\
\nabla_\theta J(\theta) &= \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a)A^{\pi_\theta}(s,a)]
\end{aligned}
\tag{2.8}
$$

However, the exact advantage function is generally inaccessible in rewal-world settings, and as a result, both the action-value and value functions must be approximated. Since the TD error δ^{π_θ} is an unbiased estimate of the advantage function,

$$
\begin{aligned}
\delta^{\pi_\theta} &= r + \gamma V^{\pi_\theta}(s') - V^{\pi_\theta}(s) \\
\mathbb{E}_{\pi_\theta}[\delta^{\pi_\theta}|s,a] &= Q^{\pi_\theta}(s,a) - V^{\pi_\theta}(s) = A^{\pi_\theta}(s,a)
\end{aligned}
\tag{2.9}
$$

we can thus use δ^{π_θ} in place of the advantage function in the policy gradient expression:

$$
\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a)\delta^{\pi_\theta}] \tag{2.10}
$$

Hence, the TD error provides a practical and efficient way to estimate the advantage function using only the value function, allowing for effective policy updates. To summarize, the policy gradient has many equivalent formulations:

$$
\begin{aligned}
\nabla_\theta J(\theta) &= \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a) v_t] && \text{REINFORCE} \\
&= \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a) Q_W(s,a)] && \text{Q Actor-Critic} \\
&= \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a) A^{\pi_\theta}(s,a)] && \text{Advantage Actor-Critic} \\
&= \mathbb{E}_{\pi_\theta}[\nabla_\theta \log \pi_\theta(s,a) \delta^{\pi_\theta}] && \text{TD Actor-Critic}
\end{aligned}
\tag{2.11}
$$

2.1.2 Proximal Policy Optimization (PPO)

The policy gradient in (2.10), used in TD Actor-Critic, can be modified using importance sampling as follows:

$$
\begin{aligned}
\nabla_\theta J(\theta) &= \mathbb{E}_{\pi_\theta}\left[\nabla_\theta \log \pi_\theta(s,a) A^{\pi_\theta}(s,a)\right] \\
&= \mathbb{E}_{\pi_{\theta_{\text{old}}}}\left[\frac{\pi_\theta(s,a)}{\pi_{\theta_{\text{old}}}(s,a)} \nabla_\theta \log \pi_\theta(s,a) A^{\pi_\theta}(s,a)\right] \\
&= \mathbb{E}_{\pi_{\theta_{\text{old}}}}\left[\frac{\nabla_\theta \pi_\theta(s,a)}{\pi_{\theta_{\text{old}}}(s,a)} A^{\pi_\theta}(s,a)\right] \\
&\approx \mathbb{E}_{\pi_{\theta_{\text{old}}}}\left[\frac{\nabla_\theta \pi_\theta(s,a)}{\pi_{\theta_{\text{old}}}(s,a)} A^{\pi_{\theta_{\text{old}}}}(s,a)\right]
\end{aligned}
\tag{2.12}
$$

While using TD errors help reduce the variance of the gradient estimates, they do not directly constrain how much the policy changes during an update. PPO is desinged to improve the current policy as much as possible using the collected data, while avoiding large policy updates that can lead to instability.

Summary

State-wise Safety in Autonomous Driving via Lagrangian-based Constrained Reinforcement Learning

Summarize your thesis.

References

1. A. Einstein, "Zur Elektrodynamik bewegter Körper. (German) [On the electrodynamics of moving bodies]," *Annalen der Physik*, vol. 322, no. 10, pp. 891–921, 1905.

Appendix A

Abbreviations

GIST Gwangju Institute of Science and Technology

EECS Electrical Engineering and Computer Science

YOLO You Only Live Once

Appendix B

More appendix

You can add more appendices.

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special thanks to...

모두들 감사합니다.